

manufacture cabin item table

No.	Type	Item	Single-material Part / Assembly	Function	Composition	Materials	Need Rationale Source / Page / Verbatim Quote	Material Source / Page / Verbatim Quote
MC-1	Chamber Structure	Chamber frame	Assembly	Provide the structural skeleton of the sealed manufacturing chamber.	Frame members, corner joints, mounting bosses, lifting/handling points.	Aluminum alloy, 304/316L stainless steel, titanium alloy if mass-critical.	NASA EBF ³ patent defines the container as having a frame. Page p.15. Quote: "a frame."	NASA patent lists aluminum, titanium and stainless steel as container materials. Page p.15. Quote: "aluminum alloys and stainless steel."
MC-2	Chamber Structure	Chamber wall panel	Single-material panel or welded assembly	Enclose the workspace and provide vacuum boundary and radiation shielding.	Wall plate, welded seams, stiffeners, port cutouts.	Aluminum alloy or 304/316L stainless steel.	NASA AIAA flight-weight EBF ³ paper states the vacuum chamber was aluminum. Page p.2. Quote: "constructed of aluminum."	Vacuum chamber technical notes identify 300-series stainless steel as common chamber material. Quote: "304L." (Lesker)
MC-3	Chamber Structure	Chamber access door	Assembly	Provide service access to the chamber interior while maintaining vacuum sealing when closed.	Door panel, hinge, latch/clamp, seal groove, handle, interlock target.	Aluminum alloy or stainless steel door; FKM seal for lower temperature; metal seal for high vacuum/hot service.	NASA EBF ³ patent includes at least one door in the sealed container. Page p.15. Quote: "at least one door."	Vacuum quick-access door example lists aluminum door frame, FKM sealing and Borofloat viewport. Quote: "Door sealing FKM." (OMICRON Technologies)
MC-4	Chamber Structure	Chamber viewport	Assembly	Provide optical access for operator viewing and camera monitoring.	Window optic, flange, seal, retaining ring, protective shutter optional.	Borosilicate glass, fused silica, sapphire, ZnSe for IR; 304L/316L flange.	NASA patent includes at least one window for operator visibility and cameras. Page p.7. Quote: "at least one window."	Vacuum viewport catalog lists glass and flange materials. Quote: "Quartz Sapphire ZnSe." (neyco.fr)
MC-5	Chamber Interface	Electron-gun mounting port	Assembly	Mechanically mount the fixed electron beam gun and allow the beam to enter the chamber.	Top port flange, bolt pattern, locating datum, seal, stiffening ring.	Aluminum or stainless chamber flange; OFHC copper gasket for CF-style port if used.	NASA AIAA EBF ³ paper states the electron beam gun is inserted through the top. Page p.3. Quote: "inserted through the top."	CF flange notes list stainless steel flange materials and copper gaskets. Quote: "304L, 316L." (Lesker)
MC-6	Chamber Interface	Wire-feeder mounting port	Assembly	Support the wire feeder near the process zone and preserve alignment to the beam.	Mounting flange, bracket interface, locating pins, seal, cable pass-through nearby.	Aluminum alloy, 304/316L stainless steel, titanium alignment pins if needed.	NASA EBF ³ patent describes the wire feed subsystem mounted to the sealed container or gun. Page p.10. Quote: "wire feed subsystem."	Material follows chamber/frame material; NASA patent supports aluminum and stainless steel. Page p.15. Quote: "stainless steel."

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MC-7	Chamber Interior	Interior sacrificial shield	Assembly	Protect chamber walls and mechanisms from spatter, condensed metal vapor, and beam- process contamination.	Removable shield panels, standoffs, screws, overlap tabs.	304/316L stainless steel sheet, aluminum sheet, titanium sheet, Inconel near hot regions.	NASA patent describes protective shielding against condensed metal vapor. Page p.10. Quote: "protective shielding."	Material is engineering selection; stainless and aluminum are supported as vacuum chamber materials. Quote: "300- series stainless steel." (Lesker)
MC-8	Chamber Interior	Build-table access and positioning- system floor interface	Assembly	Provide mechanical mounting and access for the four-axis positioning system inside the chamber.	Floor plate, tapped holes, datum features, rail mounts, access clearance.	Aluminum tooling plate, stainless steel inserts, titanium or Invar datum pins if needed.	NASA AIAA paper states the positioning system is attached to the floor. Page p.3. Quote: "attached to the floor."	Material is chamber- structure engineering; final choice depends on stiffness, thermal expansion, and vacuum compatibility.
MC-9	Chamber Operations	Chamber lighting fixture	Assembly	Provide illumination for visual inspection and camera monitoring.	LED/light source, mount, shield, cable, feedthrough connection.	Aluminum heat sink, glass lens, copper wiring, ceramic/metal feedthrough interface.	NASA patent lists lighting means within instrumentation. Page p.11. Quote: "means for lighting."	Material is lighting-vendor specific; final materials require vacuum- compatible lighting datasheet.